

DEVIXO SOLUTIONS

DATA SCIENCE INTERNSHIP PROGRAM

TASK 01 – EXPLORATORY DATA ANALYSIS

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Internship Domain: Data Science

Dataset: Supermarket Sales

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn

1. Objective

The purpose of this task is to explore a real-world supermarket transaction dataset and turn the raw records into useful business information. The analysis covers data understanding, cleaning, feature analysis, correlation analysis, visualization and practical recommendations.

2. Dataset Description

The selected dataset contains 1,000 sales transactions and 17 original columns. It includes branch and city information, customer type, gender, product line, unit price, quantity, tax, total transaction value, date, time, payment method, COGS, gross margin percentage, gross income and customer rating. The data covers January to March 2019.

Dataset source: <https://www.kaggle.com/datasets/aungpyaeap/supermarket-sales>

3. Data Cleaning Process

The dataset was checked for missing values and duplicate records before analysis. Date and time fields were converted to appropriate formats, and additional Month, Day and Hour features were created. Range checks were also applied to quantity, transaction value and customer ratings. The original dataset is relatively clean, so no artificial imputation was needed.

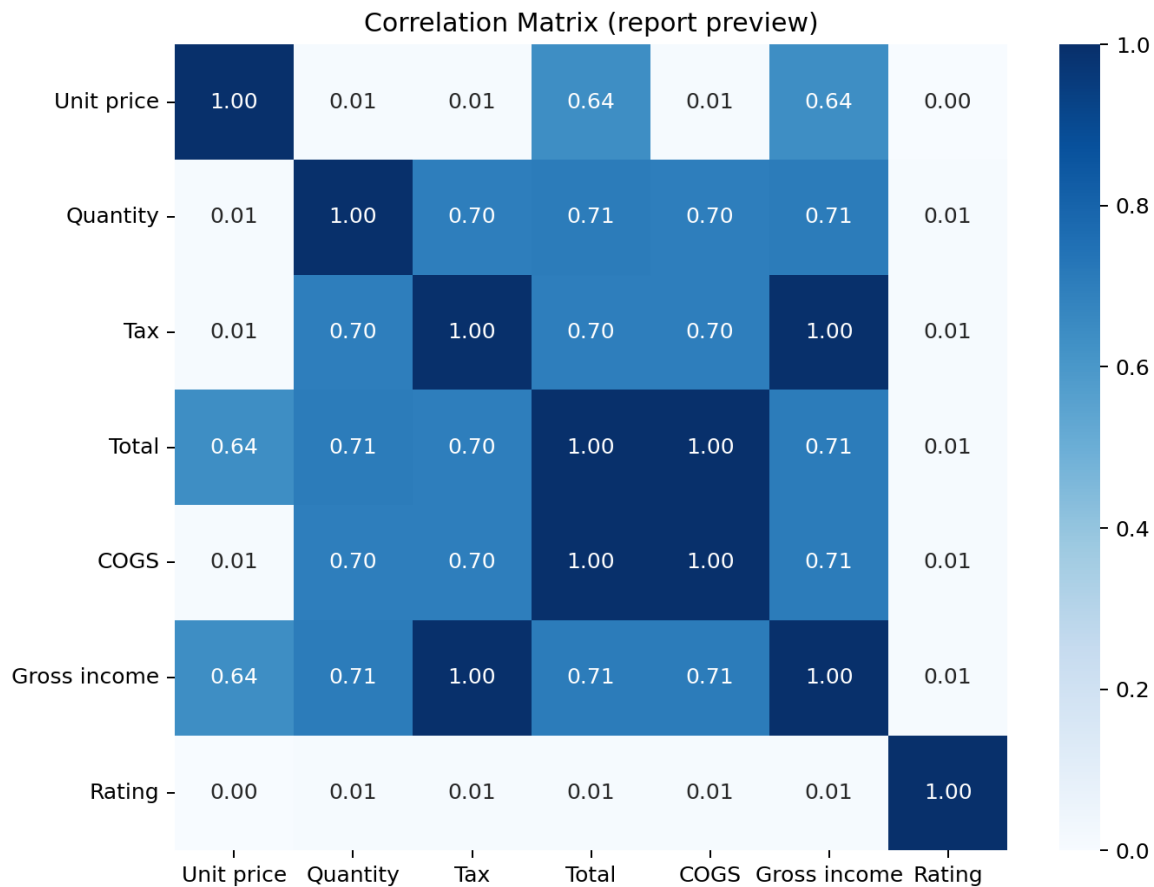
4. Feature Analysis

The average unit price is about \$55.67, the average quantity is 5.51 items per transaction, the average transaction value is about \$322.97 and the average rating is about 6.97/10. The maximum transaction value is about \$1,042.65, which shows that a small number of high-value baskets pull the mean upward.

Metric	Value
Transactions	1,000
Features	17
Average Unit Price	\$55.67
Average Quantity	5.51
Average Transaction	\$322.97
Average Gross Income	\$15.38
Average Rating	6.97 / 10
Max Transaction	\$1,042.65

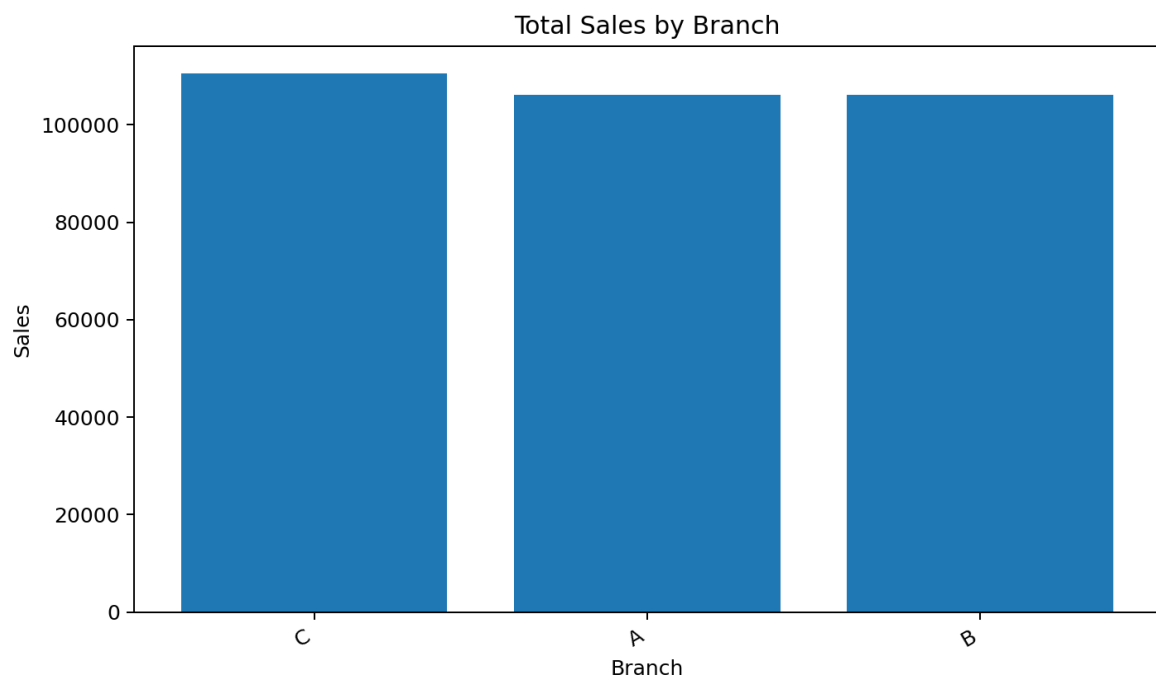
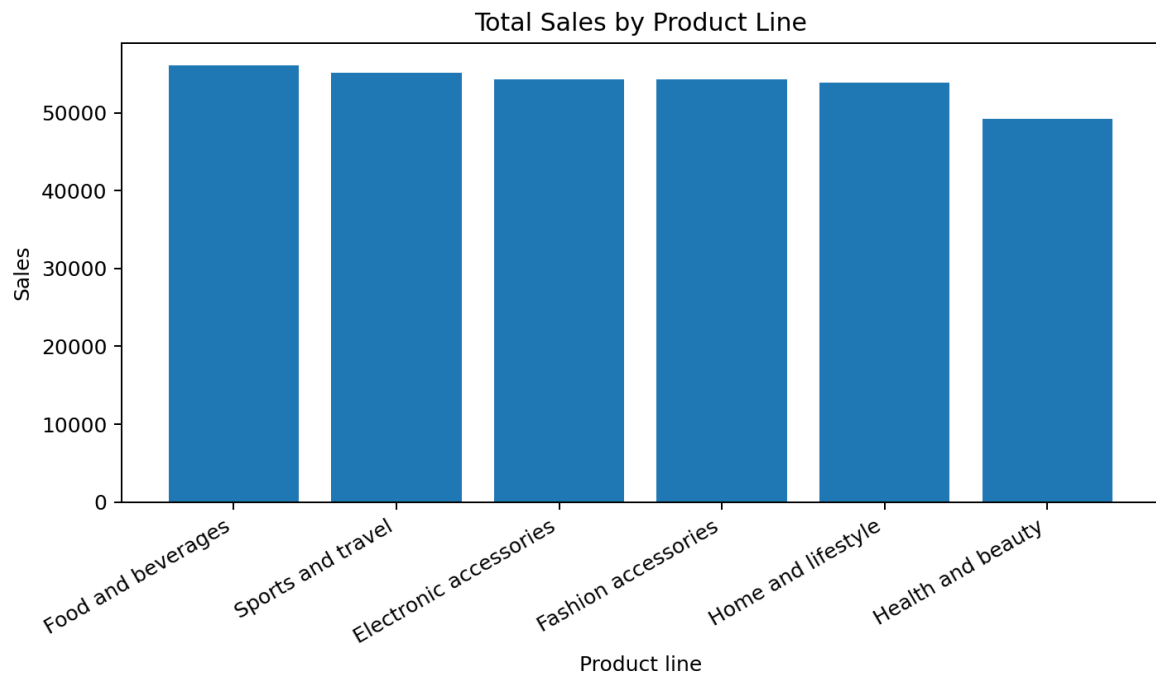
5. Correlation Analysis

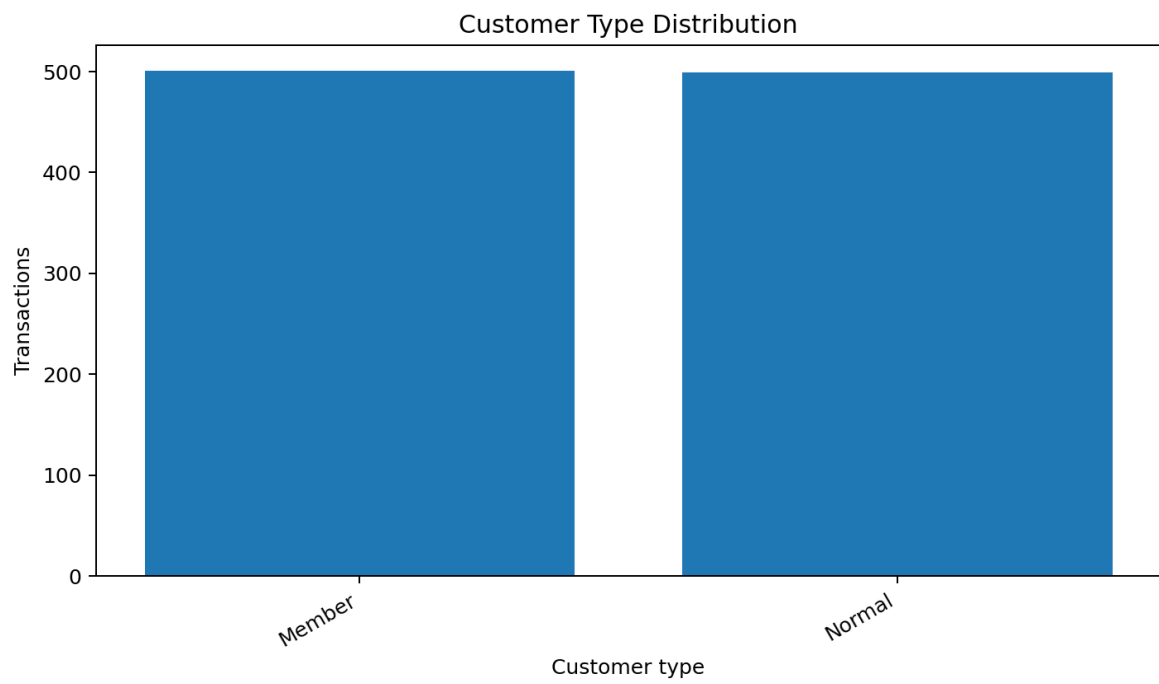
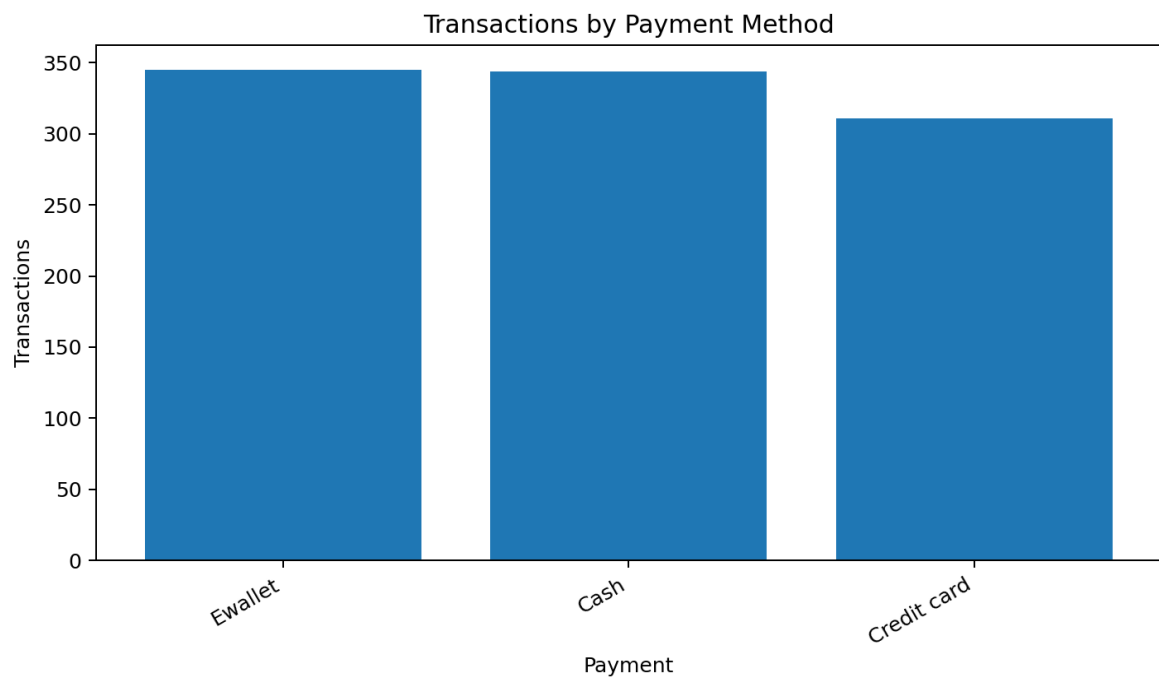
The correlation analysis shows the expected strong relationships among transaction-value variables. Quantity and total sales move together because larger baskets naturally produce larger bills. Unit price also has a positive relationship with total value. Rating has little relationship with revenue variables, suggesting that higher spending does not automatically mean a higher customer rating.

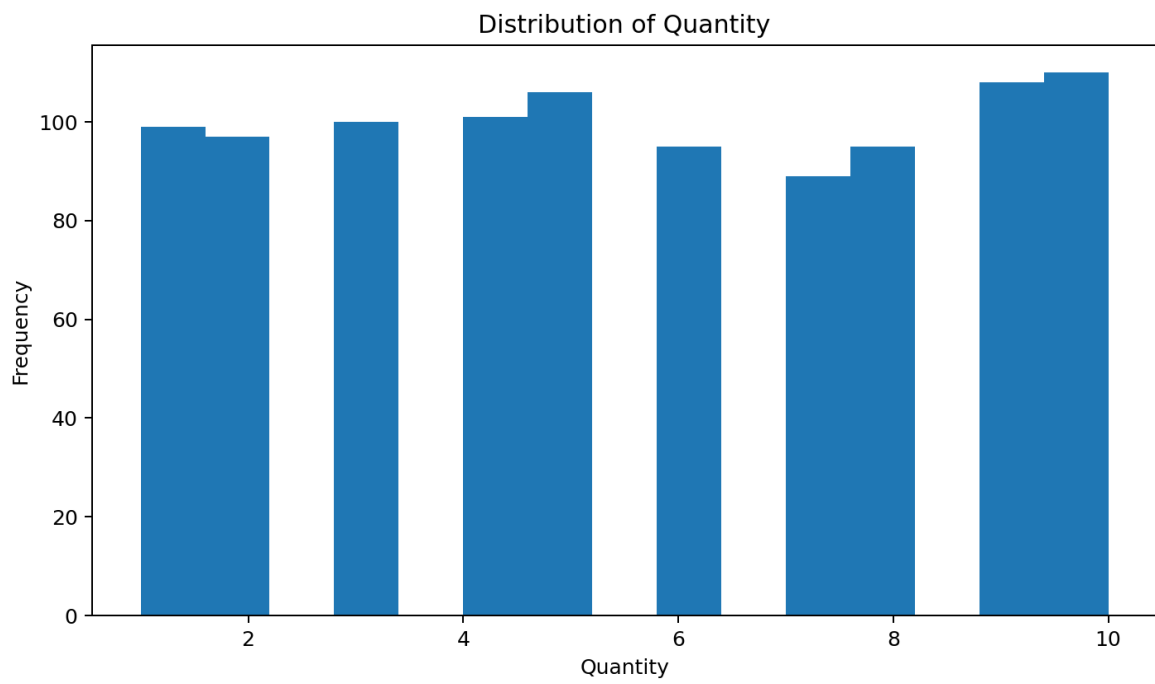
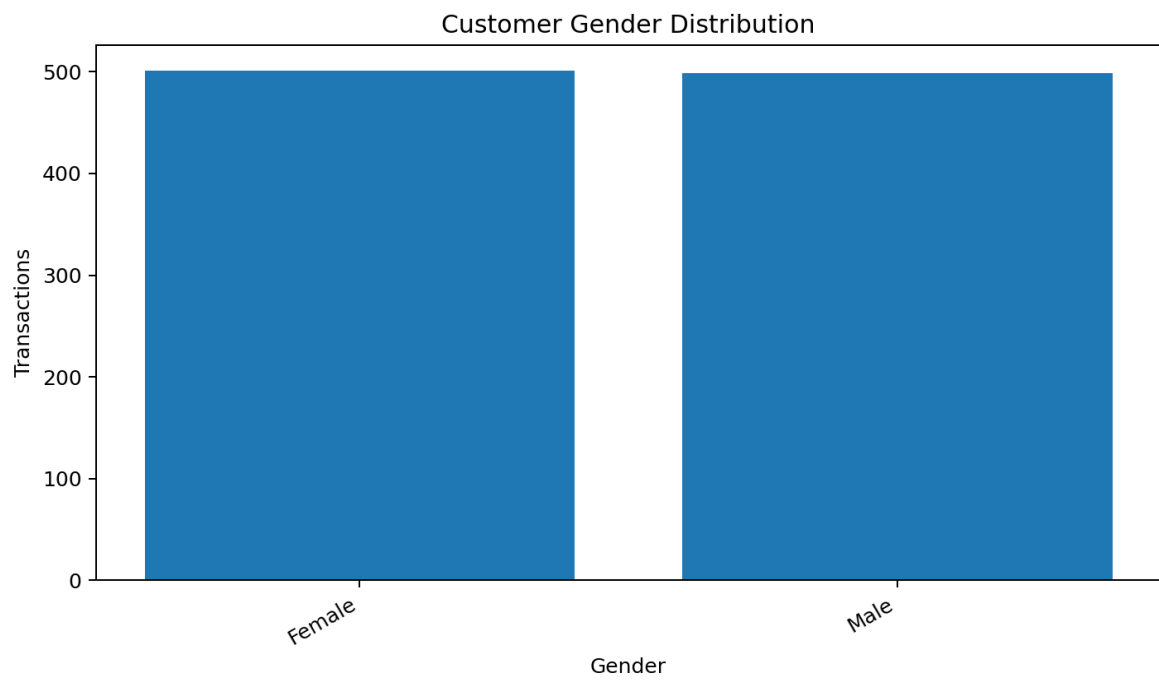


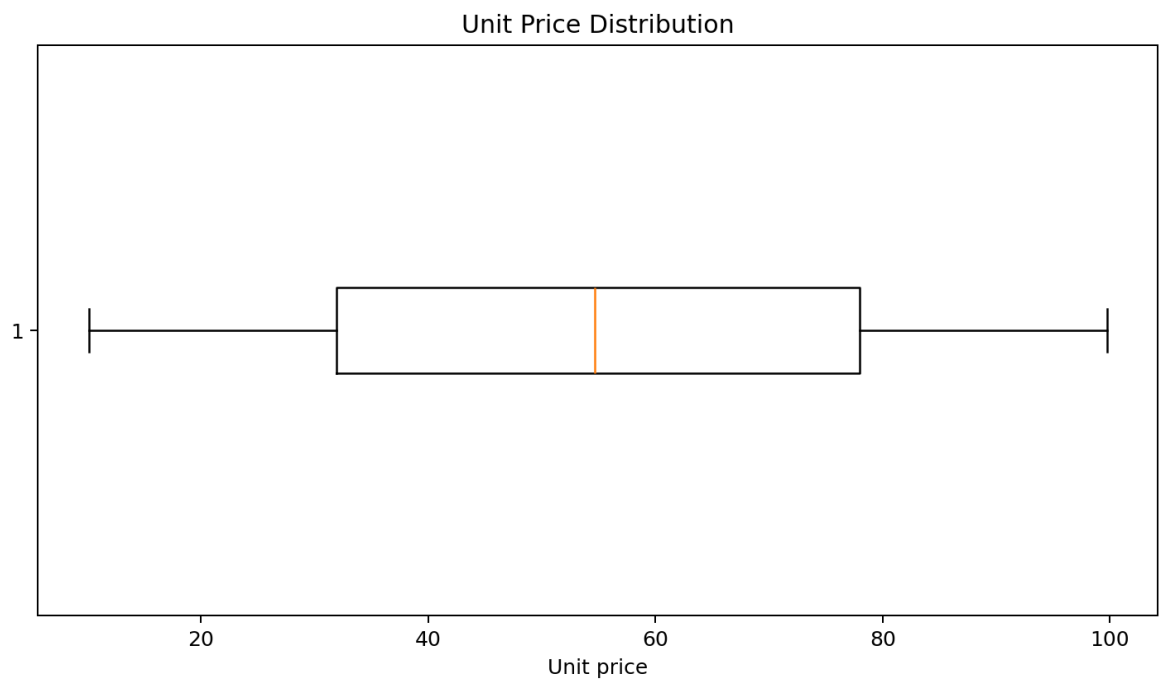
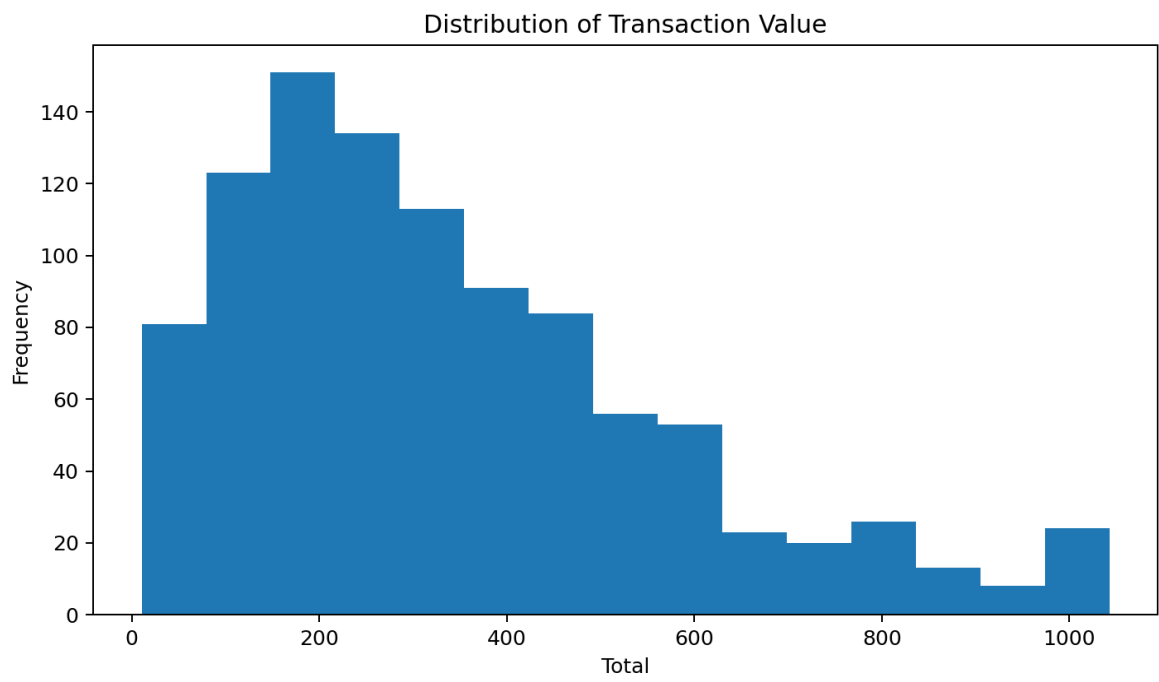
6. Visualizations

The following visuals were selected to cover product performance, branch performance, customer composition, payment behaviour, transaction distributions, timing and relationships between numerical features.

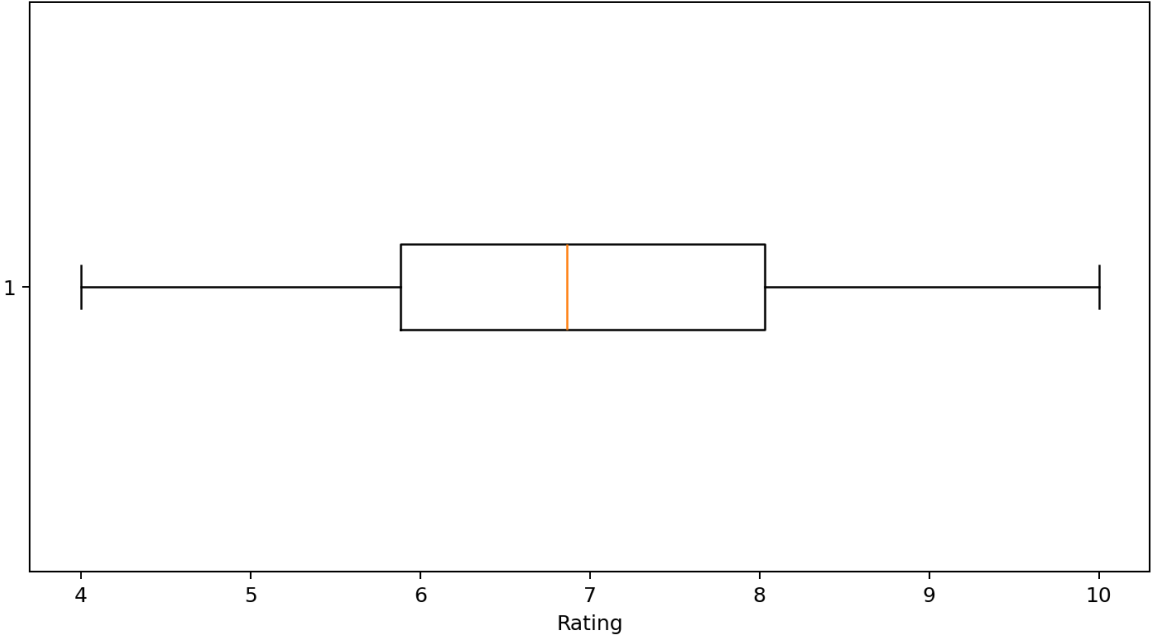




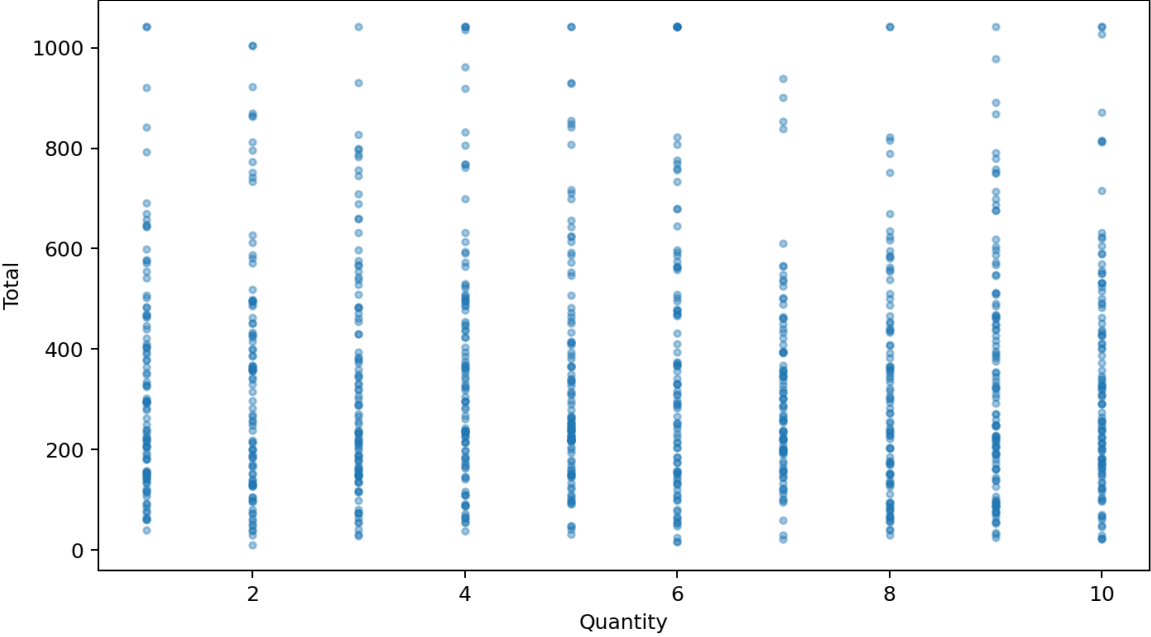


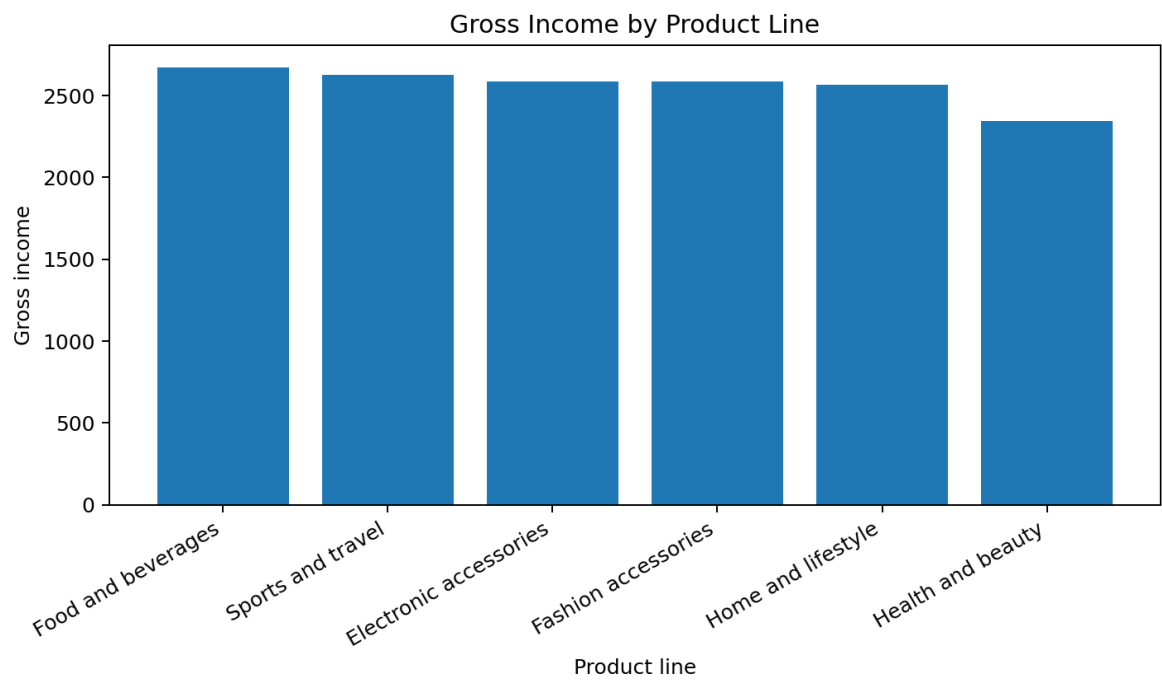
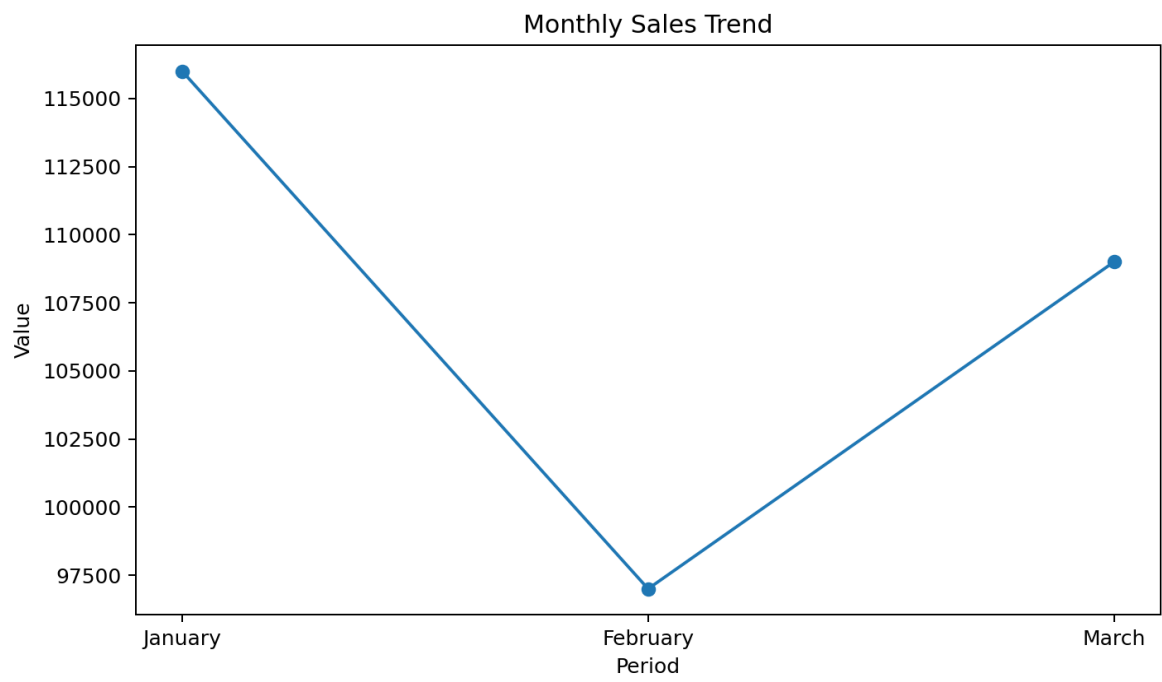


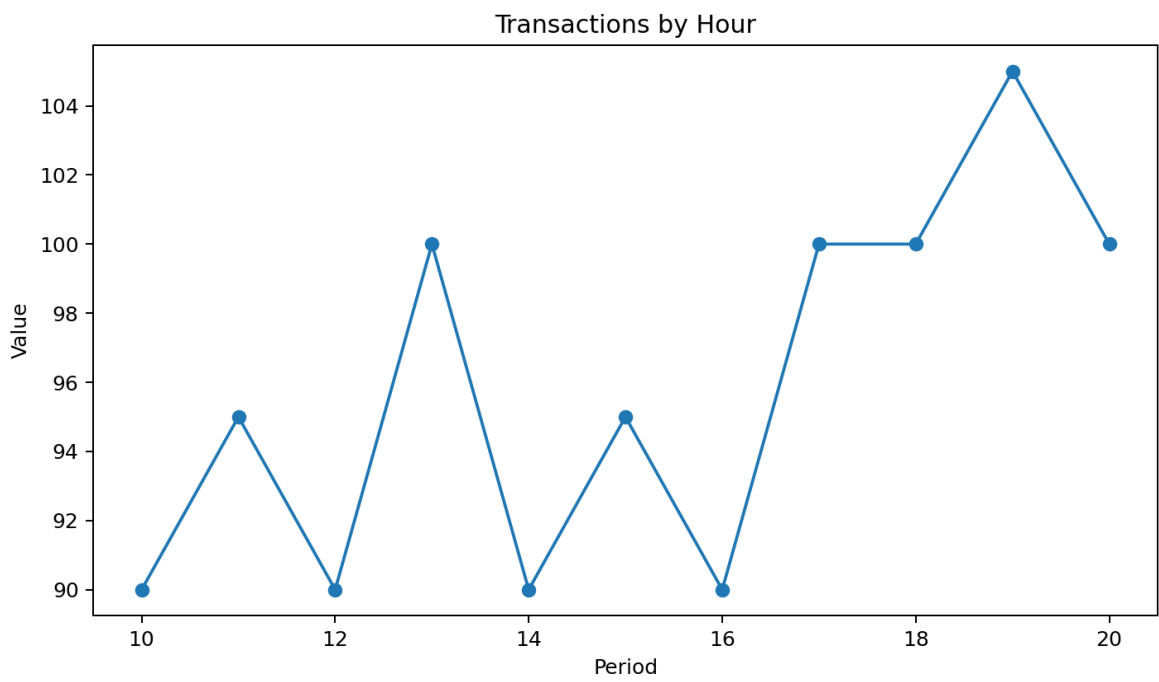
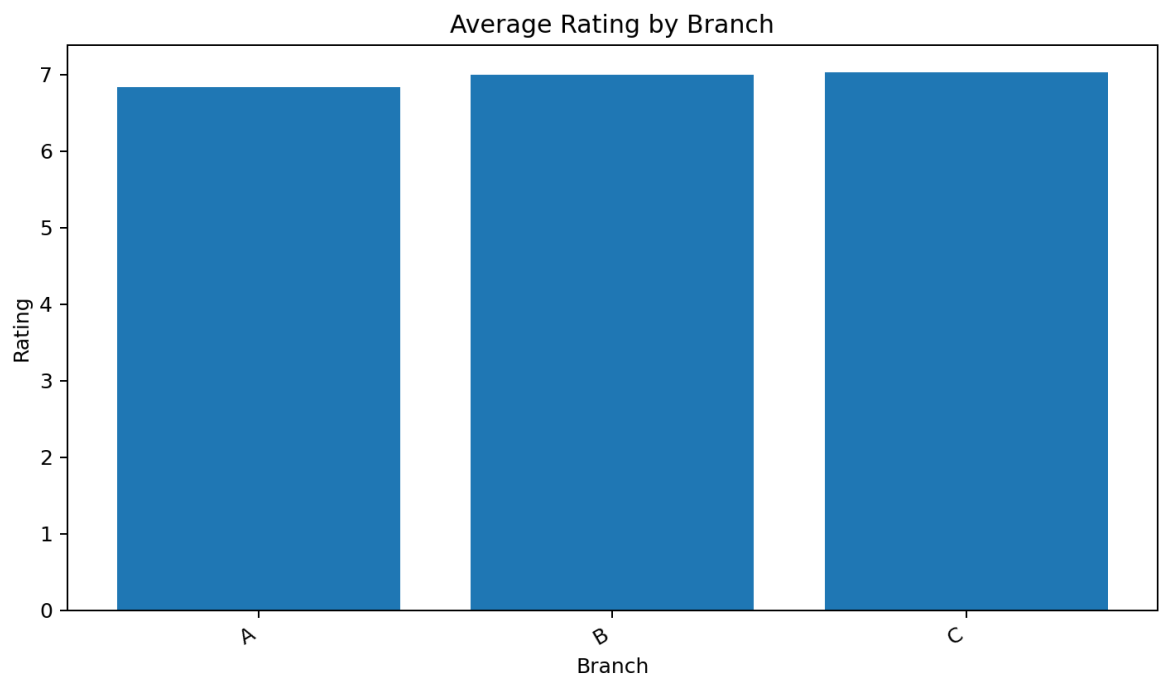
Rating Distribution



Quantity vs Total







7. Business Insights

1. Food and Beverages is one of the strongest revenue-generating product lines, so its availability should be monitored closely.
2. Health and Beauty is one of the weaker categories by revenue, making it suitable for targeted promotions rather than broad discounting.
3. Branch C / Naypyitaw is the strongest branch by total sales in this dataset.
4. Sales are reasonably balanced across the three branches, so the business is not dependent on a single location.
5. Cash, E-wallet and credit-card payments are all important, showing that customers use multiple payment options.
6. Member and normal customers contribute similar revenue, so the loyalty program has room to create stronger differentiation.
7. The average transaction is about \$322.97, while the distribution contains many smaller baskets and a smaller number of high-value purchases.
8. Quantity has a strong positive relationship with total transaction value and is one of the clearest drivers of basket size.
9. Unit price is positively related to transaction value, but price alone does not explain the whole bill because quantity also matters.
10. The average rating is around 6.97/10, which indicates acceptable customer experience but leaves room for improvement.
11. Ratings range from 4 to 10 in the dataset, with no extremely low ratings recorded.
12. Gross income is tightly connected to sales variables because the dataset uses a fixed gross margin percentage.
13. Transaction activity varies by hour, so staffing can be adjusted according to the busiest periods.
14. Monthly sales fluctuate across the quarter rather than showing a smooth upward trend.
15. Product categories are relatively close in revenue, which reduces the risk of depending on only one category.
16. High-value transactions are visible in the upper tail of the total-sales distribution and can be investigated for basket composition.
17. Branch performance should be evaluated using both sales and ratings because financial performance and customer satisfaction measure different things.
18. The strongest improvement opportunities are inventory planning, targeted promotions, loyalty engagement and time-based staffing.

8. Business Recommendations

1. Protect inventory for high-performing categories such as Food and Beverages and Sports and Travel because stock-outs in these categories can directly affect revenue.
2. Use targeted offers for weaker categories such as Health and Beauty instead of applying discounts to every product.
3. Strengthen the membership program with member-only bundles or repeat-purchase offers because member and normal customer revenue is currently close.
4. Use the hourly transaction pattern to schedule more staff during busy periods and avoid overstaffing quieter periods.
5. Investigate lower-rated branches or product lines through customer feedback on service, product availability and checkout experience.
6. Keep cash, E-wallet and credit-card payment options available because customer usage is distributed across all three.

7. Review high-value transactions to identify common product combinations and use those combinations for cross-selling or bundles.

9. Conclusion

This EDA demonstrates the full workflow from raw transaction data to business recommendations. The analysis highlights the strongest product lines and branch, customer and payment patterns, transaction size, customer ratings and time-based activity. The next stage could use the cleaned data for feature engineering and predictive modelling.

10. GitHub Repository

Repository folder prepared with notebook, report, visualizations, screenshots, README and requirements. Upload this folder to a new GitHub repository named **DEVIXO-Task01-EDA**.